

Falcons

- **Problem Statement ID – 26173**
- **Problem Statement Title:- iTantra -Indian Multilingual TTS & STT**
Aided Neural Transceiver Radio Access for low bitrate links
- **Theme-Smart Automation**
- **PS Category- Software**
- **Team ID-**
- **Team Name : Falcons**

Problem:

- iTantra -Indian Multilingual TTS & STT Aided Neural Transceiver Radio Access for low bitrate links
- Bandwidth Limit: Raw audio cannot pass through low-bitrate links (Bluetooth/Wi-Fi/Mesh) during emergencies.
- Cloud Dependency: Current STT/TTS requires active internet, high RAM, and cloud APIs unusable offline.

Our Idea:

- Problem You're Solving: Inability to communicate during network blackouts, in remote areas without internet/cellular coverage, or across language barriers.
- How It Works: Local AI converts voice to text/meaning --> Compresses it into tiny data packets --> Sends it over local device-to-device mesh--> Translates and speaks it out on the receiver's phone.
- Key Differentiator: 100% offline, privacy-first, zero cloud dependency, and uses virtually no network bandwidth.

Innovation / Uniqueness:

- Ultra-Low Bandwidth** Semantic Packetization
- Converts speech into ~50–200 byte semantic packets, **reducing voice bandwidth by 98%**.
- Zero-Infrastructure Mesh Networking**
- Uses **Wi-Fi Direct + Bluetooth Mesh** for peer-to-peer communication without cellular towers or cloud servers.
- Fully Offline** Edge AI
- Runs STT, translation, and voice synthesis entirely on-device ensuring privacy, **zero server cost**, and **low latency**.
- Real-Time Multilingual Translation**
- Enables users to **speak in their native language** and receive translated speech in their preferred language completely offline.



Proposed Solution:

1. Capture & Voice Filtering:

- 16 kHz PCM audio ingestion in 100 ms chunks.
- Silero VAD removes silent packets.

2. Speech-to-Text & Serialization:

- IndicConformer converts speech into text locally.
- Protobuf packs text and metadata into compact frames.

3. Ad-Hoc Transport:

- Wi-Fi Direct transfers frames over local TCP.
- Bluetooth RFCOMM provides fallback connectivity.

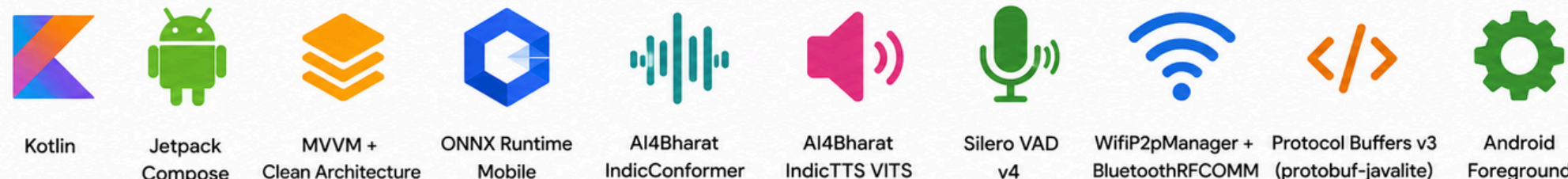
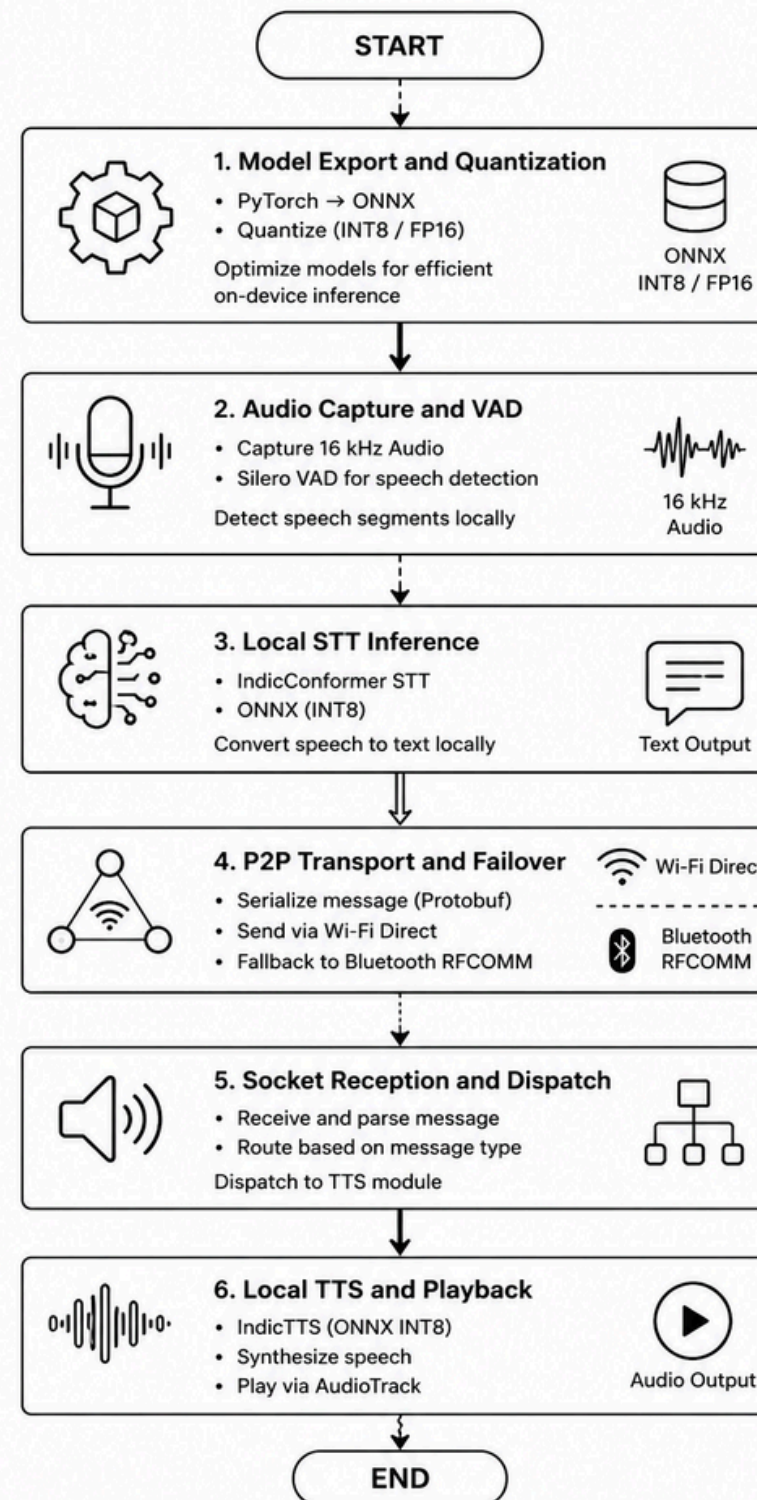
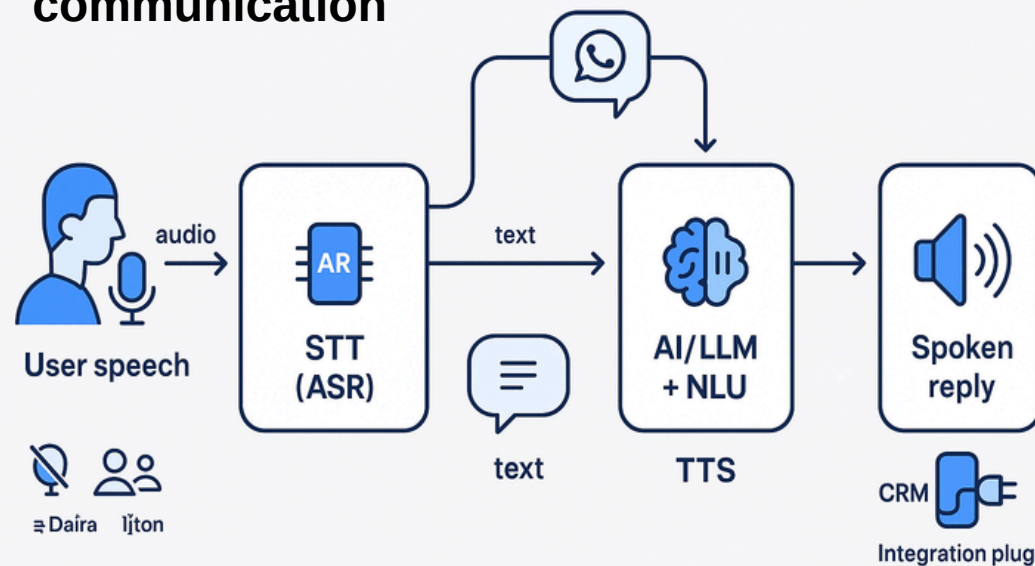
4. Ingestion & Priority Routing:

- Receiver unpacks incoming Protobuf frames.
- Data is routed by priority to the required components.

5. Local Speech Synthesis:

- IndicTTS VITS converts text into natural speech offline.
- AudioTrack handles playback with SOS alert override.

1. On-device multilingual speech recognition and processing
2. Semantic compression for ultra-low bandwidth transmission
3. Tantra Protocol enables efficient P2P communication
4. Protobuf-based framing ensures reliable data transfer
5. Wi-Fi Direct with Bluetooth fallback connectivity
6. Retrieval-LLM enhances contextual response generation
7. On-device TTS reconstructs natural human speech
8. Fully offline, privacy-preserving end-to-end communication



FEASIBILITY

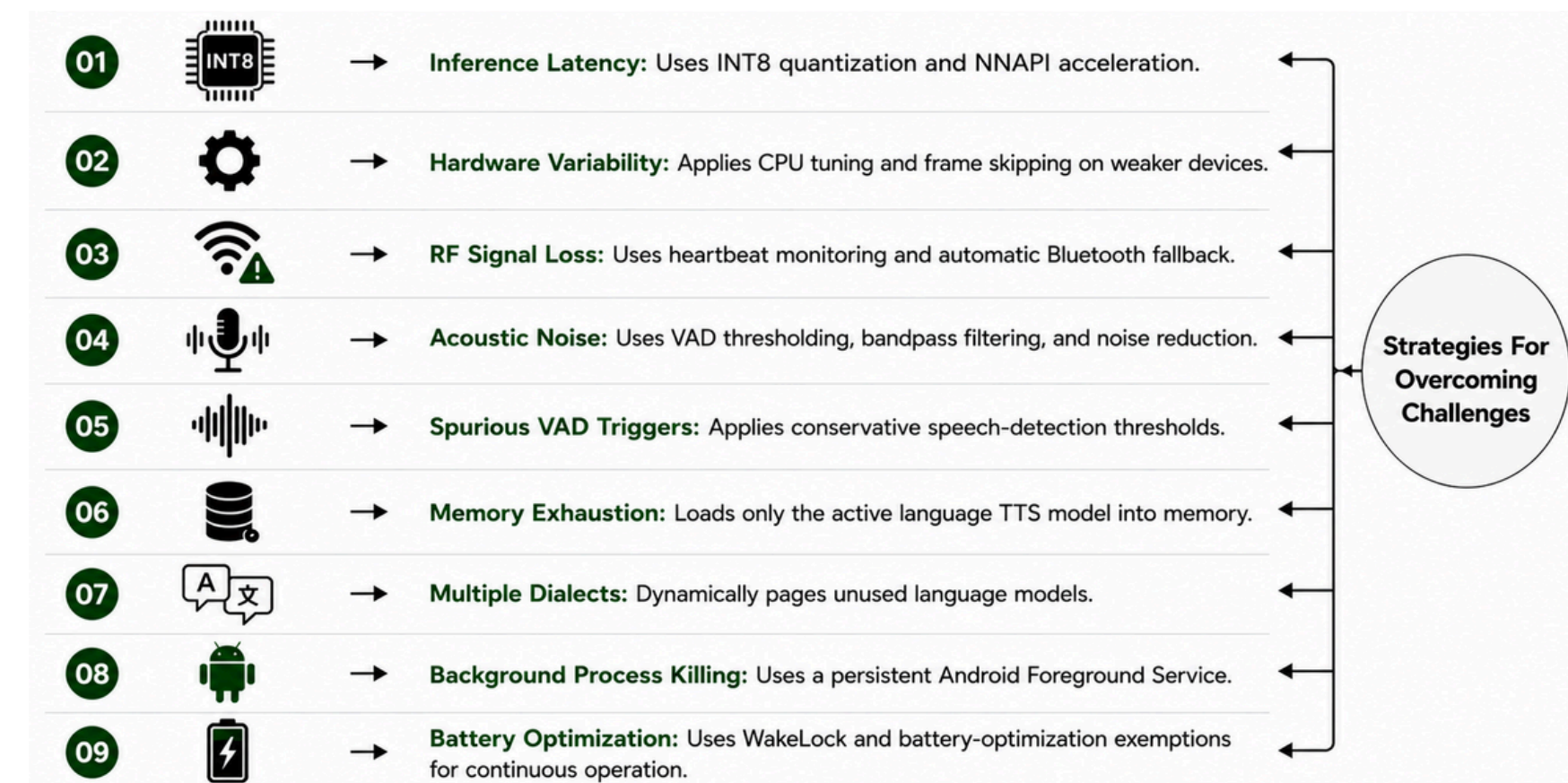
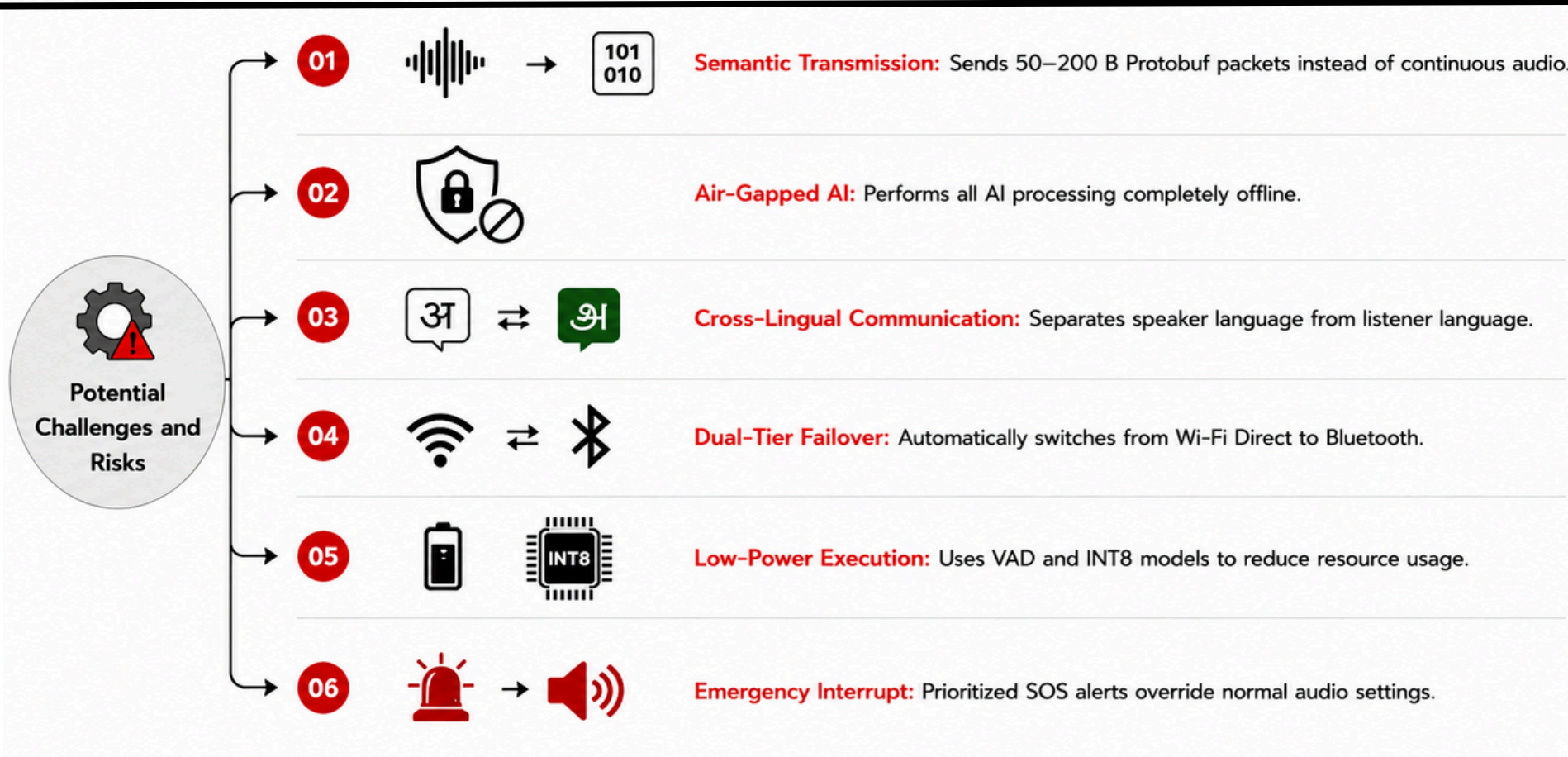
- **Hardware & Runtime:** Runs locally on mobile chipsets using ONNX Runtime Mobile and NNAPI.
- **Storage & Memory:** INT8 models require low storage and controlled RAM usage.
- **RF Connectivity:** Uses built-in Wi-Fi Direct and Bluetooth RFCOMM.
- **Open-Source Licensing:** Uses Apache 2.0 and MIT-licensed technologies.

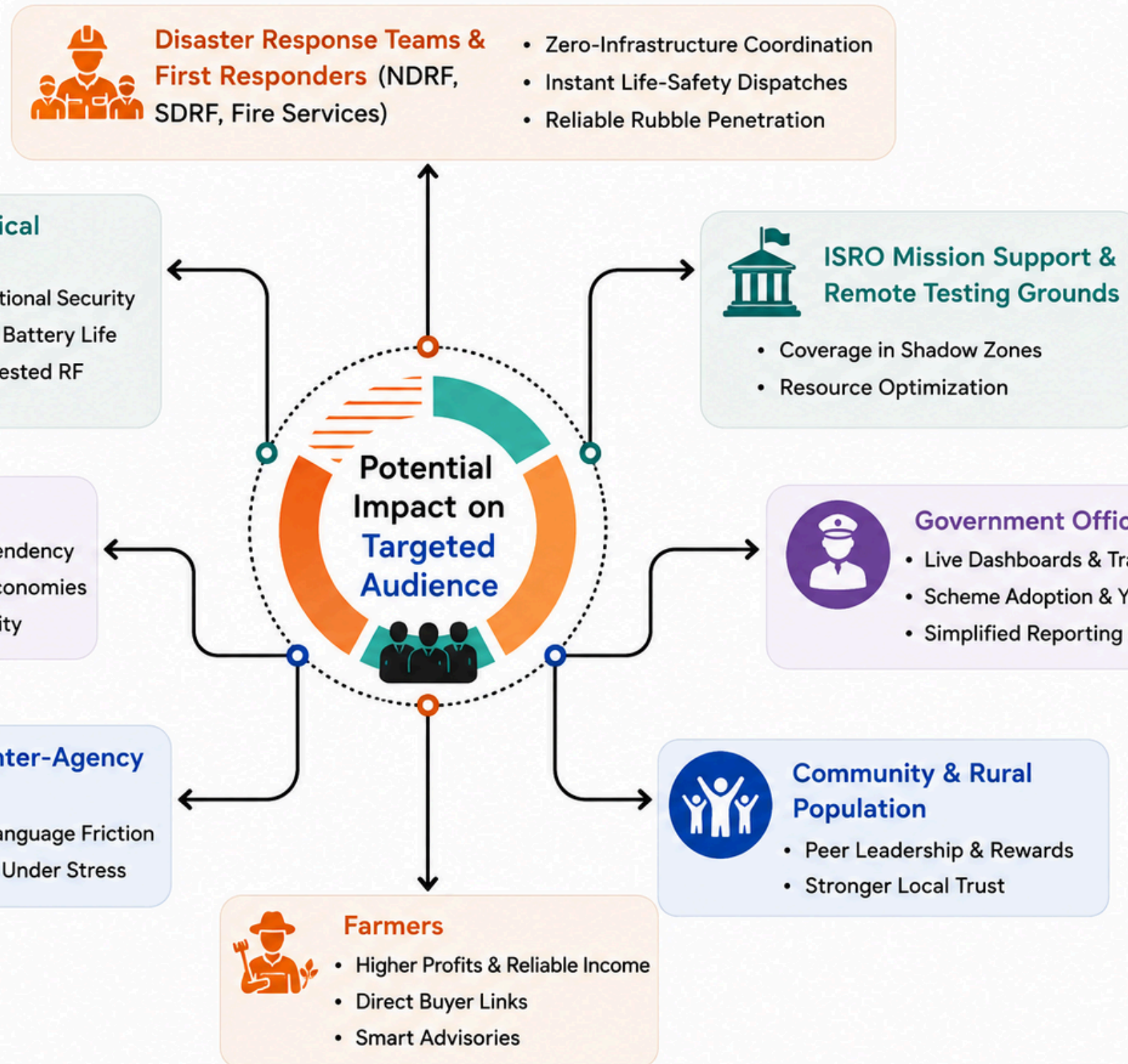
VIABILITY

- **Zero Infrastructure Cost:** No cloud, servers, or satellite subscriptions required.
- **Power Efficiency:** VAD reduces STT/TTS processing during silence.
- **Secure Offline Design:** Operates without internet or cloud dependency.
- **Inter-Agency Scalability:** Protobuf supports standardized communication between teams.

Practical Implementation

- **Zero-Setup Pairing:** Automatic device discovery and ad-hoc mesh formation.
- **Push-to-Talk Workflow:** Simple PTT interface for regional-language speech.
- **Adaptive Transport:** Automatic Wi-Fi Direct to Bluetooth switching.
- **Low-Latency Feedback:** Synthesized speech delivered in under 2.5 seconds.
- **Fail-Safe Alerts:** SOS messages use high-priority alarm playback.
- **Local Processing:** STT and TTS run directly on the device.





Benefits of the solution

01



Instant Deployment

Works without cellular, satellite, or Wi-Fi infrastructure.

02



Bandwidth Reduction

Reduces communication data requirements by over 98%.

03



Low Latency

Provides speech communication with under 2.5-second latency.

01



Multilingual Interoperability

Enables communication across different regional languages.

02



Critical Alert Delivery

Ensures priority SOS alerts receive immediate playback.

03



Reduced Cognitive Load

Converts distorted speech into clear synthesized audio.

Field & Applied Research

- Semantic Communication: Transmits 50–200 B meaning-based data instead of continuous voice, reducing bandwidth by >98%.
- Offline Multilingual AI: Enables real-time cross-language communication using on-device Indic STT/TTS without internet or cloud.
- Resilient Ad-Hoc Networking: Combines Wi-Fi Direct + Bluetooth auto-failover, keeping communication alive even when infrastructure is unavailable.

Academic & Government Sources

- **IndicSUPERB** — Speech Processing Universal Performance Benchmark for Indian Languages (AI4Bharat, 2022): Benchmark evaluations, multi-accent Indic acoustic datasets, and IndicConformer ASR model performance across 12 Indian languages.
- **Conformer** — Convolution-Augmented Transformer for Speech Recognition (Google Research, 2020): Foundational architecture paper on combining convolution and self-attention mechanisms for parameter-efficient, low-latency end-to-end ASR.
- **Towards Building Text-To-Speech Systems for the Next Billion Users (AI4Bharat / ICASSP, 2023)**: Neural acoustic modeling, vocoder optimization, and natural speech synthesis evaluations for 13 Dravidian and Indo-Aryan languages.
- **Conditional Variational Autoencoder with Adversarial Learning for End-to-End Text-to-Speech (VITS, 2021)**: Non-autoregressive neural generative architecture enabling real-time waveform synthesis directly on edge hardware.
- **Wi-Fi Alliance — Wi-Fi Peer-to-Peer (Wi-Fi Direct) Technical Specification**: Industry standard and communication protocols governing zero-infrastructure, ad-hoc device discovery and local P2P socket routing.